SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
PRACTICAL ASSIGNMENT 10

1. Student Details

Student Name	Homesh Sawarkar
PRN	202501110069
Division	A
Batch	A2
Roll No.	202501110069
Date of Submission	27/10/2026

2. Practical Details

Practical No.	10
Practical Title	Templates
Date Performed	-
Faculty Name	Khushal Khairnar

3. Problem Statement

Implement class template, function template.

4. Objectives

1. To understand the concept of templates and their importance in generic programming using C++.
2. To implement class templates and function templates to work with different data types.
3. To improve code reusability and reduce code duplication by writing type-independent programs

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Explain the concept and advantages of templates in C++.
2. Create and use class templates to store and process different data types.
3. Implement function templates to perform operations on different data types using a single function definition.

6. Theory / Concept

Templates are a feature of C++ that allows programmers to write generic code that works with different data types. They improve code reusability and reduce the need to write separate functions or classes for each data type.

Function Template: A function template defines a generic function that can operate on different data types. The compiler generates the appropriate function based on the data type used when the function is called.

Class Template: A class template allows a class to work with different data types. It is useful for creating generic data structures and classes that can store or process values of various types.

7. Algorithm / Pseudocode

START

Step 1: Include the required header files and declare a function template that accepts two values of the same data type.

Step 2: Implement the function template to compare the two values and return the larger value.

Step 3: Define a class template containing a data member and member functions to store and display a value of a generic type.

Step 4: Create objects of the class template using different data types and call the function template with different arguments.

Step 5: Display the results to demonstrate that both templates work with multiple data types.

STOP

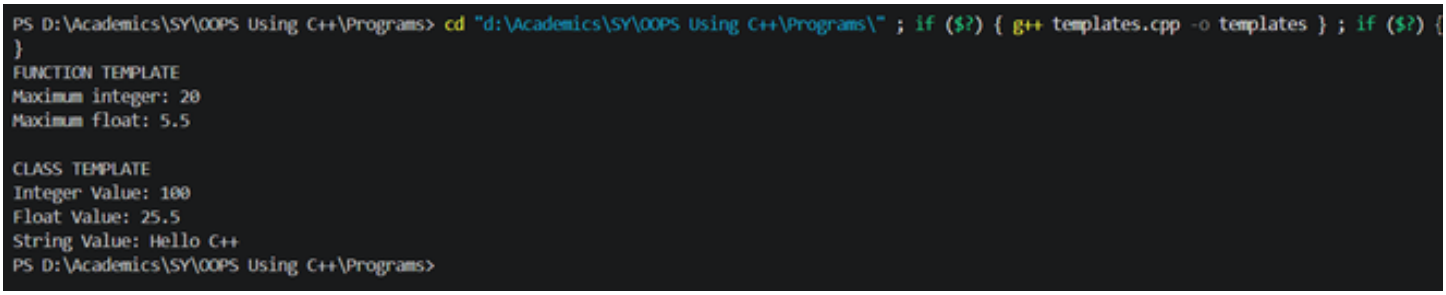
08. Technology / Tools / Libraries Used

Technology / Tool	**Purpose**
Programming Language	C++
IDE / Platform	VS Code
Libraries / Packages	-

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	https://github.com/Homeshsawarkar/Assignment-10-C-
Repository Name	Assignment 10 C++
Branch Used	main
Commit ID / Latest Commit	63d3e23236d0292ff7428a7a3f24607f96b2938d
Date of Last Commit	Oct 27, 2026

10. Results / Observations



11. Conclusion

In this practical, function templates and class templates in C++ were implemented and tested with different data types. The function template performed operations on different types of values, while the class template stored and displayed integer, floating-point, and string data. This practical helped develop an understanding of generic programming, code reusability, and type-independent code in C++.

12. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
Homesh Sawarkar	202501110069	Homesh A.S